

```
1  module datatype_tb;
2
3      bit          a_bit      = 1'b1;
4      logic        b_logic    = 1'b0;
5      logic [3:0]  c_logic4    = 4'b1111;
6
7      byte         d_byte     = -5;
8      shortint     e_short    = -100;
9      int          f_int      = 200;
10     longint       g_long     = -1000;
11
12     logic signed  [3:0] h_signed_logic ;
13
14     initial begin
15         #5ns;
16
17         h_signed_logic = 4'b1000;
18         $display("===== UNSIGNED =====");
19         $display("a_bit          = %0d", a_bit);
20         $display("b_logic        = %0d", b_logic);
21         $display("c_logic4       = %0d", c_logic4);
22
23         $display("\n===== SIGNED =====");
24         $display("d_byte         = %0d", d_byte);
25         $display("e_short        = %0d", e_short);
26         $display("f_int          = %0d", f_int);
27         $display("g_long         = %0d", g_long);
28
29         $display("\n===== SIGNED LOGIC =====");
30         $display("h_signed_logic (bin) = %b", h_signed_logic);
31         $display("h_signed_logic (dec) = %0d", h_signed_logic);
32
33         $display("\n===== MIXED OPERATION =====");
34         $display("Sum of unsigned + signed (15 + -5) = %0d", c_logic4 + d_byte);
35
36         $display("\n===== DONE =====");
37     end
38
39 endmodule
```

```

#
# run -all
# ===== UNSIGNED =====
# a_bit          = 1
# b_logic        = 0
# c_logic4       = 15
#
# ===== SIGNED =====
# d_byte         = -5
# e_short        = -100
# f_int          = 200
# g_long         = -1000
#
# ===== SIGNED LOGIC =====
# h_signed_logic (bin) = 1000
# h_signed_logic (dec) = -8
#
# ===== MIXED OPERATION =====
# Sum of unsigned + signed (15 + -5) = 10
#
# ===== DONE =====
# exit
# End time: 20:33:38 on Jul 03,2025, Elapsed time: 0:00:01
# Errors: 0, Warnings: 0
End time: 20:33:38 on Jul 03,2025, Elapsed time: 0:00:02
*** Summary ****

```